Ornstein-Uhlenbeck Mean Reversion Process Applied on S&P 500 Point-in-Time Constituents over 20 Years: Short Half-Life Against Trending Assets

One long-only Ornstein-Uhlenbeck z-score rule, with fixed settings, traded on the five S&P 500 stocks with the shortest half-life of mean reversion and on five trending stocks, re-selected every six months from 2006 to 2025 and compared with SPY as run and at equal volatility.

Serhat Girgin · QuanterLab UG (haftungsbeschränkt), Germany · contact@quanterlab.com · October 2026

ABSTRACT

The Ornstein-Uhlenbeck process is the standard model of a mean-reverting price: deviations from a long-run level decay at a rate usually expressed as a half-life. A common trading rule built on it buys a stock when the price is far below the estimated level, measured in units of the stationary standard deviation of the model (a z-score), and sells when the price has returned close to that level. We ask whether the half-life helps to decide which stocks the rule should trade.

Every six months from January 2006 to December 2025 we estimated the half-life of each S&P 500 member with a price history in our data, using the index membership of that date. One portfolio held the five stocks with the shortest half-life; the other held five trending stocks, ranked on a long half-life and a steep one-year trend. Both traded the same long-only z-score rule, with the estimation period and entry and exit levels of Avellaneda and Lee (2010), identical for every stock and fixed before the first test. Each six-month test was locked before it ran. Positions were sized in equal shares, by half Kelly and by full Kelly.

Before trading costs, the short half-life portfolio turned $1 into $2.78 and the trending portfolio into $1.90, against $5.38 for SPY. Scaled to the volatility of SPY, the short half-life portfolio returned 9.1% a year against 8.8%. In both groups about two trades in three were profitable, with the same ratio of average gain to average loss; the trades on the short half-life stocks were profitable slightly more often. Kelly sizing changed little. Measured over the six months in which they were traded, the half-lives of both groups were close to those of a random walk, so the test cannot separate the effect of the half-life from that of avoiding trending stocks. A charge of a tenth of a percent on every purchase and sale would bring the scaled portfolio below SPY.

Keywords: mean reversion; Ornstein-Uhlenbeck process; half-life; statistical arbitrage; z-score; Kelly criterion; volatility scaling; point-in-time S&P 500; walk-forward test. **JEL classification:** C22, G11, G12, G14.

Generative AI was used to draft prose from the registered record and to check the draft against it; the author reviewed and edited every sentence and takes full responsibility for the content.

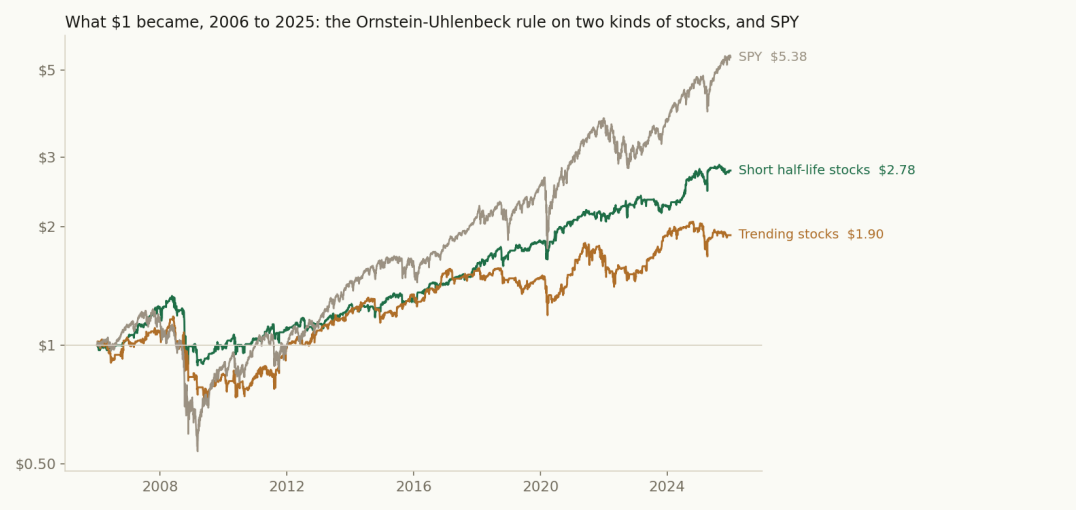


Figure 1. Growth of $1 from the first close of 2006 to the last close of 2025: the long-only Ornstein-Uhlenbeck z-score rule on the five shortest half-life stocks and on five trending stocks, re-selected every six months, with equal shares, and SPY. Logarithmic scale; price returns before trading costs.

1. INTRODUCTION

Mean reversion, the tendency of a price that has moved away from a reference level to move back toward it, is the basis of a large family of short-term trading rules. Weekly and monthly reversals in the returns of single stocks were documented by Jegadeesh (1990) and Lehmann (1990), and slower mean reversion in market prices by Poterba and Summers (1988); Lo and MacKinlay (1988) showed that weekly index returns depart from a random walk. The Ornstein-Uhlenbeck process (Uhlenbeck and Ornstein, 1930) is the simplest continuous-time model of such behaviour and the common starting point of statistical arbitrage (Elliott, van der Hoek and Malcolm, 2005; Avellaneda and Lee, 2010; Bertram, 2010).

A trader who applies the model to single stocks must first choose the stocks. The model offers its own criterion: the speed of reversion, usually reported as a half-life, the expected time for a deviation from the mean to fall by half. Avellaneda and Lee (2010) kept only residual series whose estimated reversion time was short, and Chan (2013) recommends the half-life as a screen. A stock with a short half-life should give the rule frequent and quick round trips; a stock in a persistent trend should give it entries that keep moving against it.

We test this on the S&P 500 from 2006 to 2025 with a design that leaves the trading rule unchanged and varies only the stocks it receives. Every six months the members of the index are ranked by their half-life over the preceding year; one portfolio takes the five shortest, the other five stocks with long half-lives and steep rising trends. Both trade the same z-score rule with settings fixed in advance. We report the results as they ran, at the volatility of SPY, after an arithmetic charge for trading costs and under three sizing schemes, and we check how far the selection measure persisted into the months in which the stocks were traded.

Under every sizing scheme the short half-life portfolio ended above the trending portfolio: with equal shares, $1 grew to $2.78 against $1.90, and the worst decline from a high was 33% against 41%. Its advantage came from a higher share of profitable trades (67.5% against 64.0%); the ratio of average gain to average loss was the same in both groups. As it ran, the rule held cash about three quarters of the time and grew less than SPY. Scaled to the volatility of SPY it returned 9.1% a year against 8.8% before costs, and 5.7% after a charge of 0.1% on every purchase and sale. The half-lives that defined the two groups did not persist: over the six months of trading the middle values were 12 and 15 trading days, inside the range that the same estimator gives on random walks.

2. THE ORNSTEIN-UHLENBECK PROCESS AND THE RULE

2.1 The model

In continuous time the process X follows $dX = \theta(\mu - X)\,dt + \sigma\,dW$, where μ is the long-run level, $\theta > 0$ the speed of reversion, σ the size of the random shocks and W a Brownian motion. A deviation from μ decays on average as $\exp(-\theta t)$, so its half-life is $\ln 2 / \theta$. The process settles into a stationary distribution with mean μ and standard deviation $\sigma / \sqrt{(2\theta)}$. Observed once a day it is an autoregressive process of order one, $X(t) = a + b\,X(t-1) + \varepsilon(t)$ with $b = \exp(-\theta)$, and the half-life in trading days is $-\ln 2 / \ln b$.

In the 1930 paper the equation described the velocity of a small particle in a fluid, slowed by friction and pushed by random collisions. In finance the same equation is applied to a price, to the logarithm of a price or to the spread between two prices. Figure 2 shows two simulated paths driven by the same shocks around the same level, one with a half-life of 5 days and one with 60 days: the first stays close to the level, the second drifts away for months and resembles a trend.

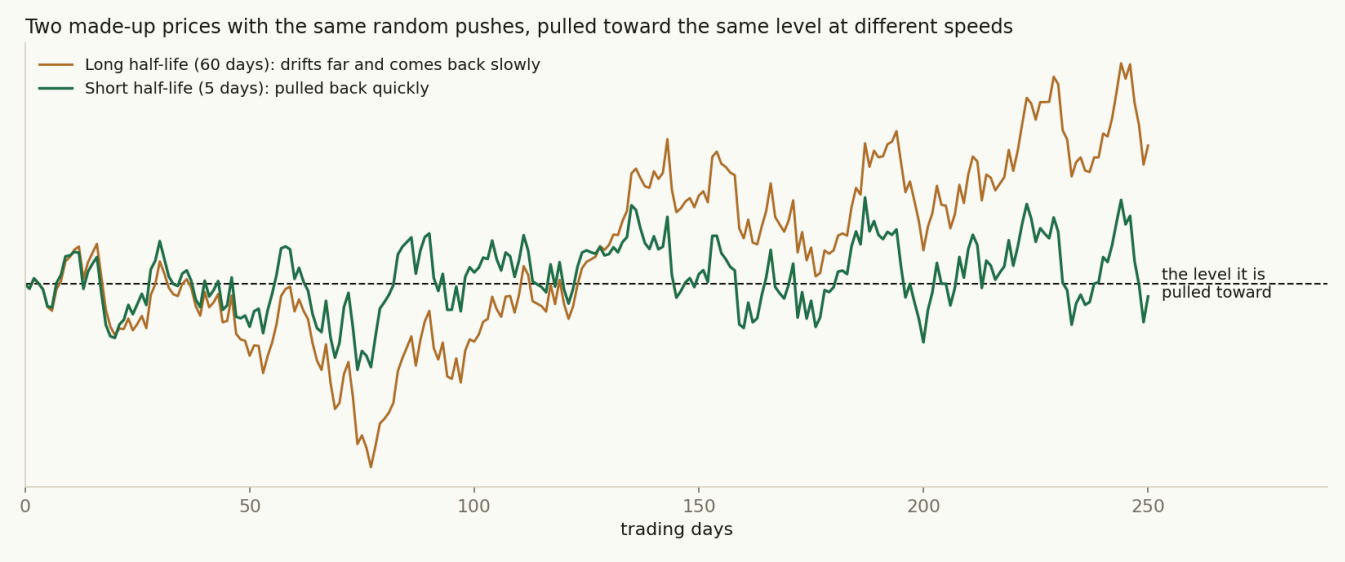


Figure 2. Two simulated Ornstein-Uhlenbeck paths driven by identical shocks around the same level (dashed), with half-lives of 5 and 60 trading days.

2.2 The trading rule

Each day the rule fits the autoregressive form to the last 60 daily closes, recovers μ and the stationary standard deviation, and computes the z-score of the latest close: its distance from μ in stationary standard deviations. A long position is opened when the z-score falls below −1.25 and closed when it rises above −0.5. The two levels and the 60-day estimation period follow Avellaneda and Lee (2010), who applied them to stock returns net of sector exchange-traded funds, long and short, and only to residuals with fast reversion. Our rule trades the price of each stock and only long.

The rule adds the stop of the platform card: a position is closed when the z-score falls below −3.5, and no new position is opened while the z-score stays below that level. Signals are evaluated at the close and filled at the same close. Figure 3 shows the rule on American Express and Apple in the second half of 2006, one from each group, with the price above and the z-score below.

2.3 The half-life as a selection measure

For the selection, the last year of daily closes of each stock is log-transformed and detrended by a least-squares line. An autoregressive regression of order one on the residual gives the coefficient b, corrected for small-sample bias by adding (1 + 3b) / n (Kendall, 1954), and the half-life −ln 2 / ln b in trading days. A corrected coefficient of 1 or more indicates no reversion; the measure is then set to a ceiling of 999 days.

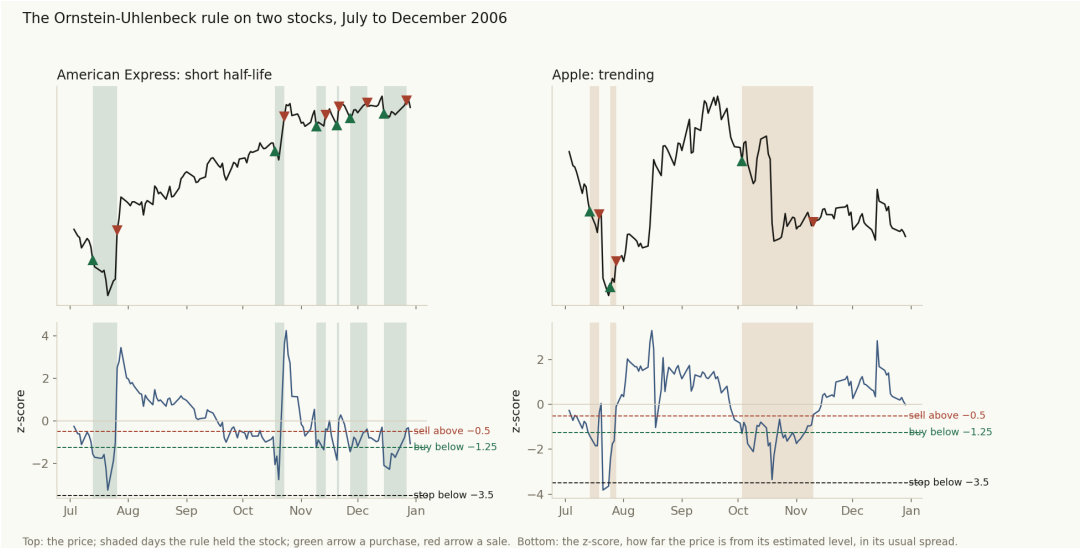


Figure 3. The rule on American Express (short half-life group, left) and Apple (trending group, right), July to December 2006. Top: price, with days in a position shaded, purchases (green) and sales (red). Bottom: the z-score with the entry level at −1.25, the exit level at −0.5 and the stop at −3.5.

3. DATA AND DESIGN

Universe. The members of the S&P 500 on each first trading day of January and July from 2006 to 2025, reconstructed from the dated history of index changes. Members without a reliable price history in our data are left out: about a third of the 2006 members and under a tenth from 2015 onward, companies that later left the index.

Selection. The short half-life portfolio takes the five members with the shortest half-life. The trending portfolio ranks every member on two measures with equal weight, a long half-life and the slope of the one-year log-price trend, and takes the top five.

Tests. Each six-month test starts in cash, trades the five stocks with the rule and ends at the last close before the next start: 40 tests per portfolio from January 2006 to December 2025. Every test was written down and locked on the platform before it ran. Each of the three sizing schemes is a separate set of 40 locked tests on identical signals; the study page carries the equal-share set.

Sizing. With equal shares each stock has a fifth of the capital while in a position. With Kelly sizing the share is f = p − (1 − p) / R, where p is the share of profitable trades and R the ratio of average gain to average loss, both from the trades of that stock under the same rule over the previous year; half Kelly uses f / 2. A stock with fewer than 12 trades or fewer than 2 losses in that year receives a fifth. A single position is capped at the whole capital and the portfolio at twice the capital. The record of the platform lists an optimizer card for both portfolios and describes the settings as optimized on a grid; that grid held one setting, 60 days and −1.25, so it selected nothing and only passes each stock's previous-year trades to the Kelly sizing.

Joining the tests. The figures chain the 40 tests from $1 at the first close of 2006 to the last close of 2025. Where a test ends before the next one starts, the day in between is carried with the positions held at the previous close, and SPY with its own closes. All returns are price returns, without dividends; cash earns nothing. Volatility is the annualized standard deviation of daily returns and the Sharpe ratio the annualized mean of daily returns over their standard deviation, without a risk-free rate.

Equal volatility. To compare portfolios that hold cash most of the time with an index that is always invested, each daily return is multiplied by k, the standard deviation of daily SPY returns over that of the portfolio, both over the full twenty years (Moreira and Muir, 2017). On days when k times the invested share exceeds the capital, the excess is borrowed; we charge 2% a year on it.

Costs. No costs are charged inside the tests. We compute them afterwards: 0.1% of the traded value for every purchase and every sale, summed over the trades and divided by twenty years, and multiplied by k for the scaled portfolio.

4. RESULTS

4.1 Short half-life against trending stocks

Table 1 and Figure 1 summarize the twenty years. With equal shares, the short half-life portfolio turned $1 into $2.78 (5.2% a year, volatility 10.3%, Sharpe ratio 0.55) and the trending portfolio into $1.90 (3.3% a year, volatility 13.3%, Sharpe ratio 0.31). The worst decline from a high was 33% for the short half-life portfolio, from May 2008 to March 2009, and 41% for the trending portfolio, from May 2008 to July 2009. Taken six months at a time, the short half-life portfolio did better in 23 of the 40 tests; the paired resampling of daily returns on the study page puts the probability that it ends ahead at 75% with equal shares and 81% with half Kelly.

Table 1. Results by portfolio and sizing, 2006 to 2025. Price returns before trading costs unless marked. Volatility and Sharpe ratio from daily returns, annualized, no risk-free rate. Costs: 0.1% of every purchase and sale.

PORTFOLIO	$1 BECAME	A YEAR	VOLATILITY	SHARPE	WORST FALL	SCALE K	AT SPY VOLATILITY	SAME, AFTER COSTS
Short half-life, equal	$2.78	5.2%	10.3%	0.55	33%	1.88	9.1%	5.7%
Short half-life, half Kelly	$2.90	5.5%	10.8%	0.55	32%	1.79	9.1%	5.8%
Short half-life, full Kelly	$3.01	5.7%	13.1%	0.49	38%	1.48	7.9%	4.9%
Trending, equal	$1.90	3.3%	13.3%	0.31	41%	1.46	4.2%	1.7%
Trending, half Kelly	$1.81	3.0%	13.3%	0.29	42%	1.46	3.8%	1.4%
Trending, full Kelly	$1.81	3.0%	13.8%	0.28	45%	1.41	3.7%	1.3%
SPY	$5.38	8.8%	19.4%	0.53	56%	1.00	8.8%	8.8%

4.2 The trades

Table 3 compares the trades: 913 for the short half-life portfolio and 844 for the trending portfolio, of which 67.5% and 64.0% were profitable. The average profitable trade gained 3.2% and 4.2%, the average losing trade lost 4.7% and 6.3%, so the ratio of average gain to average loss was 0.67 in both groups. The larger gains and losses on the trending stocks reflect their larger price swings, an annualized volatility of 34% against 25% for the middle stock of each group.

The difference between the groups is therefore the share of profitable trades. Expressed per unit of the average loss, the expected result of a trade was 0.13 for the short half-life stocks and 0.07 for the trending stocks, almost twice as much. Over the six months in which they were held, the middle trending stock rose 5.9% and the middle short half-life stock 2.4%; the rule, which waits for a price well below its mean, kept little of that rise. Each stock was in a position on about a quarter of the trading days.

4.3 Against SPY

As it ran, the short half-life portfolio grew less than SPY, $2.78 against $5.38, because about three quarters of the capital was in cash on an average day. Its worst decline was 33% against 56% for SPY, from October 2007 to March 2009, and in the calendar year 2008 it lost 20.2% against 38.3%. In the 10 six-month tests in which SPY fell, the short half-life portfolio did better than SPY 9 times; in the 30 tests in which SPY rose, 7 times.

Scaled to the volatility of SPY (k = 1.88), the short half-life portfolio turned $1 into $5.75, 9.1% a year against 8.8% for SPY, with a worst decline of 55% against 56% (Figure 4). On an average day the scaled portfolio had 48% of its capital in positions. On 16% of the days, when three or more of the five stocks were in a position at once, the scaled positions exceeded the capital, by 5% of the capital on an average day and by up to 88%; borrowing at 2% a year would cost about 0.1 of a percentage point a year. The trending portfolio, scaled the same way (k = 1.46), returned 4.2% a year.

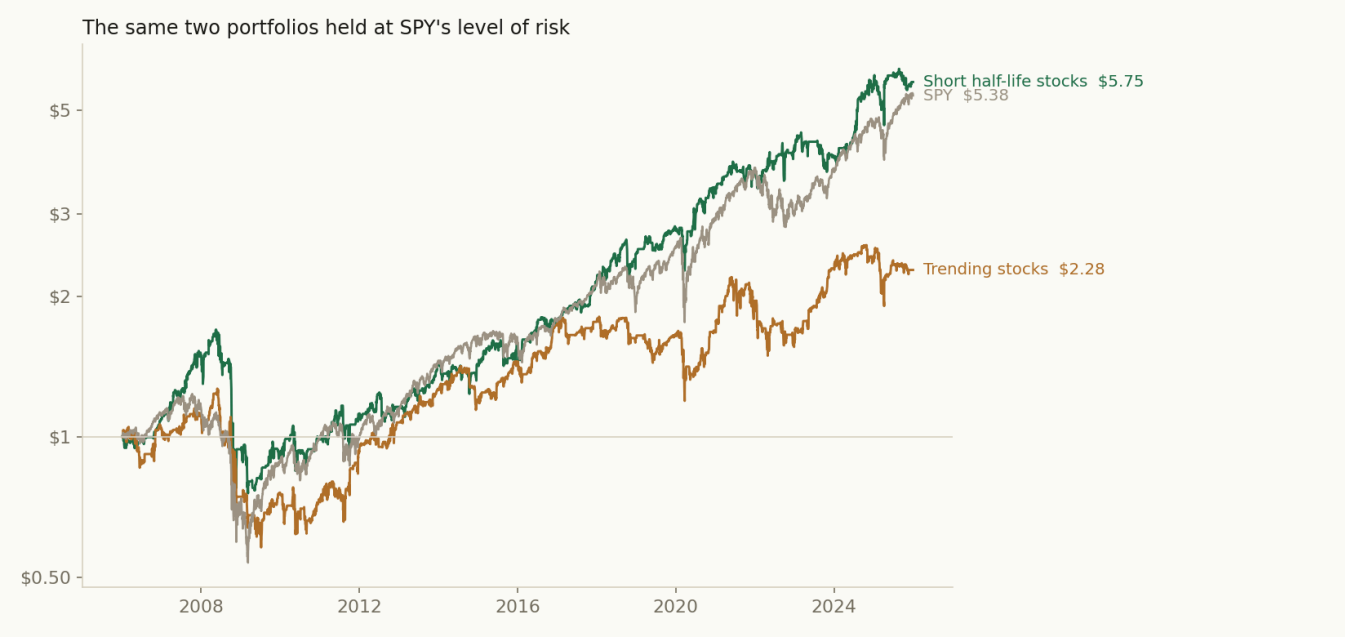


Figure 4. Growth of $1 for the two portfolios scaled to the volatility of SPY over 2006 to 2025 (k = 1.88 for the short half-life portfolio and 1.46 for the trending portfolio), and SPY. Logarithmic scale; price returns before trading costs.

Table 2. Calendar-year returns. Equal shares, price returns before trading costs. Each year runs from the last close of the year before; 2006 from its first close. The scaled column uses k = 1.88.

YEAR	SHORT HALF-LIFE	TRENDING	SPY	SHORT HALF-LIFE AT SPY VOLATILITY
2006	+4.9%	+3.3%	+11.8%	+9.2%
2007	+18.8%	+4.6%	+3.2%	+37.6%
2008	−20.2%	−23.1%	−38.3%	−37.4%
2009	−1.2%	+1.5%	+23.5%	−3.3%
2010	+5.4%	−2.0%	+12.8%	+8.8%
2011	+7.1%	+19.2%	−0.2%	+12.9%
2012	+2.3%	+10.7%	+13.5%	+3.8%
2013	+11.6%	+11.7%	+29.7%	+22.6%
2014	+0.6%	−3.2%	+11.3%	+0.7%
2015	+5.8%	+13.3%	−0.8%	+10.7%
2016	+5.8%	+13.9%	+9.6%	+10.6%
2017	+10.9%	+1.4%	+19.4%	+21.2%
2018	+8.7%	−5.1%	−6.3%	+15.9%
2019	+7.2%	+1.3%	+28.8%	+13.8%
2020	+11.3%	+2.1%	+16.2%	+20.5%
2021	+5.6%	+10.6%	+27.0%	+10.4%
2022	+7.0%	−12.3%	−19.5%	+12.6%
2023	−2.3%	+29.0%	+24.3%	−4.7%
2024	+19.4%	+2.6%	+23.3%	+38.8%
2025	+2.8%	−2.3%	+16.4%	+3.6%
A year	5.2%	3.3%	8.8%	9.1%

Table 3. Trades and stocks. Equal shares. A trade runs from a purchase to the next sale. Middle values are medians across the stocks held in each six-month test.

MEASURE	SHORT HALF-LIFE	TRENDING
Trades, 2006 to 2025	913	844
Profitable trades	67.5%	64.0%
Average profitable trade	+3.2%	+4.2%
Average losing trade	−4.7%	−6.3%
Ratio of average gain to average loss	0.67	0.67
Expected trade per unit of average loss	0.13	0.07
Worst trade	−42.9%	−56.0%
Middle stock volatility while held	25%	34%
Middle stock price change over six months	+2.4%	+5.9%
Trades per stock in six months	4.6	4.2

4.4 Trading costs

At 0.1% of the traded value for each purchase and each sale, the trades of the short half-life portfolio cost 1.8 percentage points a year as it ran and 3.4 at the volatility of SPY, where it would then return 5.7% a year against 8.8% for SPY. The scaled portfolio matches SPY only at a cost of about 0.01% per purchase and per sale. Table 1 shows every sizing after costs.

4.5 Kelly sizing

Half Kelly turned $1 into $2.90 on the short half-life stocks and full Kelly into $3.01, against $2.78 with equal shares. Full Kelly raised the volatility to 13.1% and lowered the Sharpe ratio to 0.49; at the volatility of SPY it returned 7.9% a year, below equal shares. On the trending stocks both Kelly versions ended slightly below equal shares. The formula sized only 6% of the short half-life picks and 3% of the trending picks; the rest had fewer than 12 trades or 2 losses in the previous year and

received a fifth. Full Kelly held more than the capital on 2.6% of the days, at most 165% of it.

4.6 Did the half-life persist?

Table 4 asks whether the selection measure persisted. At selection, the middle short half-life stock had a half-life between 4.1 and 8.8 trading days in every test, and the middle trending stock sat at the ceiling of 999 days in 21 of the 40 tests. Re-estimated over the six months in which the stocks were traded, the middle values were 12.5 days for the short half-life stocks and 14.6 days for the trending stocks.

The same estimator applied to simulated random walks, which have no reversion at all, gives a median of 16.0 days over six months (interquartile range 8.7 to 44.2), and, over one year, a middle value of 4.6 to 7.8 days for the five shortest out of pools of 330 or 500 walks. The short half-lives at selection are therefore close to what chance alone produces in a pool of that size, and over the trading months neither group differed clearly from a random walk.

Table 4. Half-life at selection, while traded, and on random walks. Half-life in trading days from the estimator of Section 2.3. Random walks: Gaussian daily steps, 40 draws of each pool for one year, 2,000 walks for six months, fixed seed.

SERIES	HALF-LIFE
Short half-life stocks at selection, middle value per test	4.1 to 8.8
Trending stocks at selection, middle value per test	ceiling of 999 in 21 of 40 tests
Short half-life stocks while traded, middle value	12.5
Trending stocks while traded, middle value	14.6
Random walks, one year, five shortest of 330, middle value	4.9 to 7.8
Random walks, one year, five shortest of 500, middle value	4.6 to 7.1
Random walks, six months, median (interquartile range)	16.0 (8.7 to 44.2)

5. DISCUSSION

The same Ornstein-Uhlenbeck rule produced better results on the stocks with the shortest one-year half-life than on trending stocks, under every sizing scheme and with smaller declines. The mechanism visible in the trades is a higher share of profitable round trips at an unchanged ratio of gain to loss. The test does not show that the half-life itself caused the difference: the selection measure did not persist, and the two portfolios differ in two respects, the half-life and the trend tilt of the trending portfolio. A comparison with randomly chosen stocks would separate the two and is the natural next test.

As a portfolio, the rule spends most of its time in cash, which explains both its lower growth than SPY as it ran and its smaller loss in the 2008 crash. At the volatility of SPY its return before costs was slightly higher than that of SPY, with occasional borrowing. The margin is small relative to plausible trading costs, which matters for any implementation.

The Kelly results follow from the trade counts. A rule that trades a stock about nine times a year gives too short a record for a stable estimate of p and R, and where the formula did produce a size it was large enough to raise the volatility of the portfolio without raising its growth in proportion.

6. LIMITATIONS

The sample has 40 six-month tests. A small number of years account for much of the difference between the scaled portfolio and SPY (Table 2).

Returns are price returns for both portfolios and for SPY, and cash earns nothing. SPY paid dividends of about 2% a year over the period, which would widen its lead as the portfolios ran; interest on the idle cash would work in the other direction.

Fills at the signal close assume execution at about the closing price. A next-day fill would change the entry and exit prices, more likely against a rule that buys after declines.

The scaling factor k is computed over the full period and is not available in advance; the borrowing rate of 2% a year is an assumption.

Members without a reliable price history cannot be selected. They include companies that failed or were acquired, on which a long-only reversion rule would probably have lost money, so both portfolios are likely flattered.

One parameter setting was used for the rule, one for the selection and one portfolio size. Other settings would give other results.

7. REPRODUCING THE STUDY

Every block of these tests is a card in the platform's catalog: the S&P 500 as it stood on each date and a price loader; the half-life card and, for the trending portfolio, the trend-slope card, combined by a ranking card and cut to five; the Ornstein-Uhlenbeck z-score card with the levels above and its own stop; the optimizer card set to a single value; and the six-month forward test with the chosen sizing. The circuits of the study can be opened, changed and run again on quanterlab.com.

8. DATA AND DISCLOSURE

Daily closing prices come from a licensed commercial data provider; returns are price returns, without dividends. The author is the founder of QuanterLab, whose platform ran the tests. The results describe how fixed rules behaved on historical data and are not investment advice.

Table 5. The stocks selected most often. Number of the 40 six-month tests in which each stock was among the five (3 or more).

SHORT HALF-LIFE	TESTS	TRENDING	TESTS
NiSource	4	AMD	6
TJX	3	NVIDIA	6
ExxonMobil	3	Apple	4
Cencora	3	Meta	4
Mastercard	3	Hasbro	3
Dominion Energy	3	Archer-Daniels-Midland	3
Aptiv	3	Southwest Airlines	3
FirstEnergy	3	Booking Holdings	3

References

Avellaneda, M., & Lee, J.-H. (2010). Statistical arbitrage in the U.S. equities market. Quantitative Finance, 10(7), 761–782. https://doi.org/10.1080/14697680903124632

Bertram, W. K. (2010). Analytic solutions for optimal statistical arbitrage trading. Physica A: Statistical Mechanics and its Applications, 389(11), 2234–2243. https://doi.org/10.1016/j.physa.2010.01.045

Chan, E. P. (2013). Algorithmic trading: Winning strategies and their rationale. Wiley.

Elliott, R. J., van der Hoek, J., & Malcolm, W. P. (2005). Pairs trading. Quantitative Finance, 5(3), 271–276. https://doi.org/10.1080/14697680500149370

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. Journal of Finance, 45(3), 881–898. https://doi.org/10.1111/j.1540-6261.1990.tb05110.x

Kelly, J. L. (1956). A new interpretation of information rate. Bell System Technical Journal, 35(4), 917–926. https://doi.org/10.1002/j.1538-7305.1956.tb03809.x

Kendall, M. G. (1954). Note on bias in the estimation of autocorrelation. Biometrika, 41(3/4), 403–404. https://doi.org/10.1093/biomet/41.3-4.403

Lehmann, B. N. (1990). Fads, martingales, and market efficiency. Quarterly Journal of Economics, 105(1), 1–28. https://doi.org/10.2307/2937816

Lo, A. W., & MacKinlay, A. C. (1988). Stock market prices do not follow random walks: Evidence from a simple specification test. Review of Financial Studies, 1(1), 41–66. https://doi.org/10.1093/rfs/1.1.41

Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. Journal of Finance, 72(4), 1611–1644. https://doi.org/10.1111/jofi.12513

Poterba, J. M., & Summers, L. H. (1988). Mean reversion in stock prices: Evidence and implications. Journal of Financial Economics, 22(1), 27–59. https://doi.org/10.1016/0304-405X(88)90021-9

Thorp, E. O. (2006). The Kelly criterion in blackjack, sports betting and the stock market. In S. A. Zenios & W. T. Ziemba (Eds.), Handbook of asset and liability management (Vol. 1, pp. 385–428). Elsevier. https://doi.org/10.1016/S1872-0978(06)01009-X

Uhlenbeck, G. E., & Ornstein, L. S. (1930). On the theory of the Brownian motion. Physical Review, 36(5), 823–841. https://doi.org/10.1103/PhysRev.36.823